RILEM
Proceedings
PRO 136

Edited by
Ioanna Papayianni & Jan Válek

Proceedings of the Workshop “Lime Based Materials for Repairing Historic Structures”



23 Novel Techniques for the Assessment of Injectability & Environmental Resistance of Injection Grouts for Mosaic Conservation

Francesco Santoro De Vico
Department of Mineralogy and Petrology, University of Granada,
Fuentenueva s/n, 18071 Granada, Spain
francesco.santoro.geologo@gmail.com

Maria Chiara Dalconi, Enrico Garbin, Gilberto Artioli
Department of Geosciences and CIRCe Centre, University of Padova, Italy

Maurizio Bellotto & Gilberto Artioli
Opigeo S.r.l., Padova, Italy

Abstract

In the past two decades, numerous commercial injection grouts have emerged in the market, serving as popular choices among conservators worldwide for restoring and stabilizing architectural surfaces. These grouts offer a convenient alternative to custom formulations. However, specific standardized test methods to evaluate injectability and flowability during the conservation of plasters, wall paintings, and mosaics, as well as their resistance to saline attack, have not been developed for this class of injection grouts. This study aimed to characterize five commercially available injection grouts by adapting the sand column test [1] (EN 1771 - Determination of injectability by the sand column test). The proposed adaptation allowed the assessment of grouts' ability to fill capillary networks within various granular materials, both in dry and pre-wetted states, under a constant pressure condition. Simultaneously, a systematic rheometric characterization of the grouts was performed at different water-to-solid ratios. To comprehensively understand the flowability behaviour of the grouts, the rheometric measurements were supplemented by grain size distribution profiles, imaging of the infilled porous structure, mechanical tests, and evaluation of the grouted samples' resistance to sulphate attack. The combined results from rheometric and complementary tests provided valuable insights into the flow properties of the available grouts. Additionally, these findings offer guidance for formulating improved grout compositions for mosaic conservation.

1 Introduction

Injection grouting involves the introduction of a fluid suspension behind a wall painting, plaster, or mosaic with the purpose of filling cracks and voids and re-establishing adhesion with the substrate. This technique is aimed at stabilizing architectural surfaces by introducing a bulked fluid adhesive material into the delaminated area. Grout, the material used for injection grouting, consists of binding materials mixed with an appropriate dispersing medium, possibly including fine aggregate and/or filler, and admixtures, forming a fluid yet stable injectable mixture.

The layers requiring re-adhesion may vary in thickness, ranging from very thin renders and plasters (some millimeters) to more substantial sections in mosaics (usually one or two centimeters).

Grouting is applied for in situ stabilization of architectural surfaces. This practice of conservation, first reported in the 1980s [2,3] and over the years, it has become an established conservation practice, serving as an alternative to detaching wall paintings and mosaics [4-8].

The selection of grouting materials must meet various requirements, including desired working properties, performance characteristics, and material availability.

Compatibility with the original material is crucial, encompassing physical, chemical, and mechanical aspects [9]. Often, this criterion entails using a binder similar to the original material [10, 11]. Key desired properties of grouts include good injectability [9-19], absence of sedimentation/bleeding/segregation of components [10,13-16, 20-23], and excellent durability against sulphates [12, 24, 25].

Formulating grouts is primarily an empirical process. Although commercial ready-mix grouts are available, frequently prescribed due to their ease of preparation, detailed technical information is often lacking. This lack of clarity can lead to decisions based on marketing, cost, and local availability. Commercial grouts guarantee a greater uniformity in properties and a better flow control. Furthermore, field conditions may lead to adaptations in usage conditions, deviating from the proportions suggested by the manufacturer. In this study, we compared the performance of five commercial grouts, analysing injectability, rheological properties, mechanical properties, and resistance to sulphate corrosion. We optimized the water-to-solid ratio (w/s) based on rheological measurements to enhance injectability.

2 Materials and Methods

This study compares five different binders, which are listed in Table 1 along with their suggested nominal minimum and maximum water-to-solid ratios (w/s). Among these binders, four are commercial products, while the last one, Bio Aedilitia, is a specially formulated NHL (natural hydraulic lime) grout provided by Opificio Bio Aedilitia. The Bio Aedilitia grout is classified as NHL5 "pure chaux" and is intended to be used with a w/b ratio in the range of 0.3-0.35, based on the specifications of other commercial products from the same company.

To replicate the injection tests performed in Biçer-Simsir & Rainer [1], two types of crushed brick were used, with diameters of 1.5-3.0 mm and 3.0-5.0 mm, respectively. The crushed brick samples were sieved, and fractions with sizes between two and four millimetres were carefully selected to closely match the desired size distribution curve for preparing the materials used in the injection tests.

2.1 Mixing Process

Mixing of the grouts was performed with an overhead mixer for three minutes at 500 rpm. To exclude any possible source of variability, the same mixing procedure was used for each material.

All the grouts have been prepared for the injectability test initially at w/s 0.8, and later at an optimized w/s deduced from the rheometric tests. The rheometric tests have been performed first at w/s 0.8 and at later at different w/s (see table 1) in the injectability optimization process.

Table 1
Suggested w/s of the selected grouts, initial w/s used for the injection tests and w/s used for the rheometric optimization of flow.

Grout product	Nominal min w/s	Nominal max w/s	Initial w/s	Optimization range *
Ledan TB1	0.8	1.5	0.8	0.8 - 1.0
Ledan TA1	0.8	1.6	0.8	0.8 - 1.0
PLM-1	0.3	0.8	0.8	0.4 - 0.8
PLM-SM	0.3	0.8	0.8	0.4 - 0.8
Bio Aedilitia	0.3	0.5	0.8	0.3 - 0.8

3 Analytical Techniques

XRPD was used to characterize the phase composition of the raw materials. XRPD measurements were performed using a X'Pert Pro diffractometer (PANalytical, Almelo, the Netherlands) equipped with a Co anode X-ray tube (40 kV, 40 mA), Bragg-Brentano HD optical module and X'Celerator detector. The grout powders were mixed with 10 wt% ZnO, used as an internal standard. The diffraction patterns were collected in the 3° – 85° 2θ range, with 0.017° 2θ virtual step size, counting an equivalent time of 100s per step. Rietveld quantitative phase analysis was performed using the Topas v2.1 program. The amount of amorphous phase present was estimated by applying the internal standard method.

The spectrometer used to perform the FTIR analysis was a Thermo-Nicolet IS10 coupled to a Continuum microscope. The measurements were performed in transmission in the range between 4000 – 400 cm-1.

The particle size distribution (PSD) of the grouts was measured by laser granulometry. The knowledge of PSD contributes to the understanding of the differences experimentally observed between the flow curves of the grouts. The granulometer used is a Bettersizer SD in dry mode, dispersing the dry powder in air by passing the air jet in a convergent-divergent Venturi nozzle. The granulometric distribution is calculated by modelling the sample scattering through mathematical models. In the present case the Mie theory was used.

The fluid injection grouts were tested by measuring the injectability with a protocol derived from EN 1771 [1] and by measuring the flow curve with a rotational rheometer. The hardened grouts were characterized by measuring the unconfined compressive strength and the indirect tensile strength (splitting). In addition, the durability against sulphate attack of the hardened grouts was examined.

The Hydraulic grout injection test used is an adaptation of the sand column test (EN 1771-Determination of injectability by the sand column test). The grout was injected under a constant pressure of 10 psi (approx. 0.75 bar) into a column, which consists of a transparent rigid plastic tube with an internal diameter of 22.00 ± 0.05 mm and a wall thickness of approx. 1.75 mm; the length of the tube used was 390 mm. The transparent tube was filled with crushed bricks with the particle size distribution discussed above. Following the protocol devised in [1] the crushed bricks used for the injection test were heated for 24 hours in an oven at 105°C and subsequently cooled for 24 more hours to reach room temperature. The column was mounted vertically, and the grout was injected from the bottom of the tube. The distance at which the grout penetrates the column, and the corresponding time was used to draw a curve to provide an empirical measure of injectability. As explained in the following in some experiments the crushed bricks were prewetted. To classify the injectability aptitude, a classification by three categories was used: EASY (E) if the column was filled and 20 mL of excess grout was collected at 1 mL/min minimum in a graduated cylinder after the column; FEASIBLE (F) if the column was filled but the overflow of 1 mL/min was not reached and DIFFICULT (D) if the injected mortar stopped and did not fill the column or the rise in the column was slower than 30 mm/min before filling the column. If the column was not filled the final height and time (Dh-t) were recorded. When the tube was filled, rheometric tests were carried out on the overflown, to check for changes during infiltration.

After curing for 28 days, cylindrical samples with a diameter of 22 mm and a height of 44 mm were prepared by cutting the columns used for the injection test. Unconfined compression tests were carried out for each sample. Following the standards UNI EN 12390-3 (Compressive strength of specimens) and UNI EN 12390-1 (Dimensional conformity for specimens and formwork) before being tested the samples were capped with a CEM 52.5R cement paste with w/s of 0.5. The indirect tensile strength was also measured on the injection grouts (splitting test).

To investigate salt resistance, samples with and without crushed bricks were analysed by saline immersion cycling. The samples were initially dried at 105°C. Then the samples were weighed twice: one hour after drying and after immersion in a 6 wt% NA-2SO4 solution for two hours. After being weighed this second time the samples were dried again for 21 hours at 105°C. This cycle was performed for 25 days. At each weighing a photographic documentation of the progressive deterioration was recorded. If the material remained intact after 25 cycles, further mechanical tests were carried out.

The viscosity of the different grouts used for the injection tests were measured with a stress-controlled Rheometer Anton Paar MCR92 equipped with a plate-plate geometry, 25 mm in diameter. The temperature was controlled to 20±0.1 °C, and evaporation of water was avoided thanks to a solvent trap. The flow curves were measured by a triangular procedure with logarithmic ramps of shear rate from 10−4 s−1 to 10 s−1 and back allowing 5 s measuring time per point. Rheometric analysis was used as a method to identify the visco-plastic characteristics of materials. Only the ramp down data was analysed.

To visualize the porous structure infiltrated by the grouts, polished column sections were analysed for each sample. A blue pigment was placed on the sections to fill the voids present on the sample surface. High-quality photographs and scans were then taken. These images were analysed with ImageJ and Multispec software to perform classifications to distinguish the areas containing crushed bricks, grout, and porosity.

4 Results

4.1 XRD Analysis

The phases quantified by the Rietveld analysis are reported in Table 2. We can distinguish the materials into three groups. The Ledan grouts are characterized by calcite filler and differ in the filler/binder ratio, higher in Ledan TB1. The PLM grouts have quartz filler, and a characteristic small quantity of talc and magnesite as additional phases. Grout PLM-1 has a higher percentage of talc and includes portlandite. Grout PLM-SM has a higher C3S content and C3S/C2S ratio. Grout Bio Aedilitia has a different binder composition and calcite filler.

Table 2
Mineralogical phases present in the tested grouts. The e.s.d. are given in brackets.

Phases	Ledan TB1	Ledan TA1	PLM-1	PLM-SM	Bio Aedilitia
Calcite	76.2(4)	57.7(3)	4.0(1)	5.2(1)	56.0(5)
C3S	9.1(1)	22.5(1)	7.6(2)	26.6(2)	12.5(1)
C2S	2.3(1)	5.0(1)	11.6(1)	4.3(1)	12.6(1)
C3A	-	1.7(1)	0.5(7)	1.9 (1)	-
C4AF	-	-	-	-	5.2(1)
Gypsum	-	1.0(1)	0.4(1)	1.4(1)	-
Bassanite	0.6(1)	0.5(1)	-	1.0(1)	-
Quartz	0.6(1)	0.5(1)	44.6(1)	45.3(3)	2.9(3)
Magnesite	-	-	1.0(1)	0.3(1)	-
Talc 2M	-	-	1.8(1)	0.3(1)	-
Portlandite	-	-	17.0(1)	0.7(1)	-
Palygorskite	-	-	0.4(1)	1.6(1)	-
Amorphous	10.6(5)	11.1(1)	11.1(1)	11.4(7)	10.8(9)

4.2 FTIR Analysis

The phases quantified by the Rietveld analysis are reported in Table 2. We can distinguish the materials into three groups. The Ledan grouts are characterized by calcite filler and differ in the filler/binder ratio, higher in Ledan TB1. The PLM grouts have quartz filler, and a characteristic small quantity of talc and magnesite as additional phases. Grout PLM-1 has a higher percentage of talc and includes portlandite. Grout PLM-SM has a higher C3S content and C3S/C2S ratio. Grout Bio Aedilitia has a different binder composition and calcite filler.

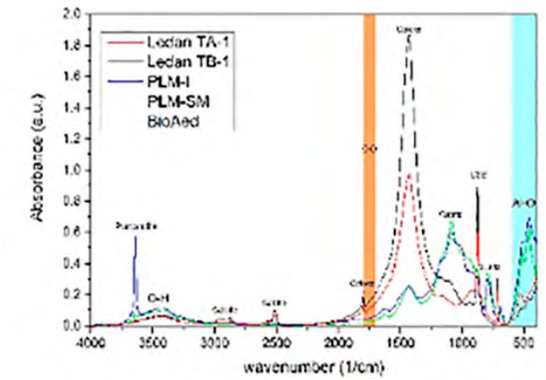


Fig. 1
FT-IR analysis of the grout binders. Highlighted in orange and blue are the vibrational regions of C-O and Al-O respectively.

4.3 Granulometry

The measured particle size curves (Figure 2) show in all samples two main particle distribution modes: one at 2–3 μm and one at 10 μm. The Ledan grouts have a higher content of fine material than the PLM and Bio Aedilitia grouts. The small particle size shoulder in the PLM samples is possibly due to the talc/palygorskite addition.

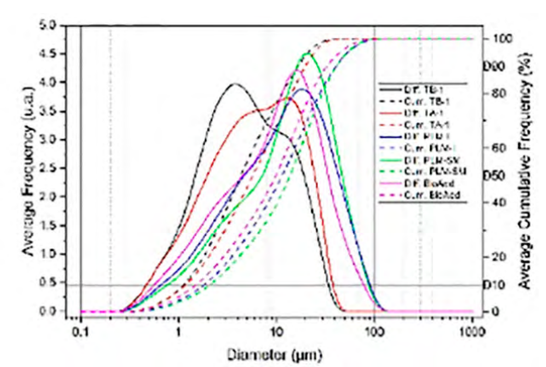
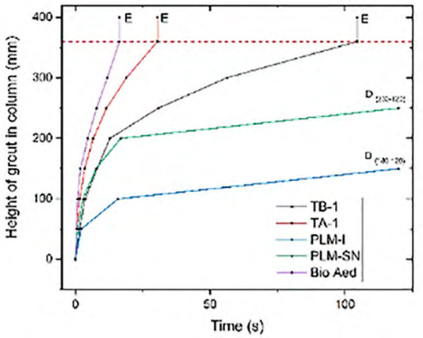


Fig. 2
Granulometric curves of the dry binder powders. The dotted line represents the cumulative analysis and the solid line the differential calculation of grain size.

4.4 Injection Test

Many parameters can influence the results of the grout injection testing: the particle size of the grout, the w/s, the mixing procedure, the injection pressure, the porosity of the particle bed and the water absorption of the material. Also bleeding or segregation phenomena occurring in the grout slurry cause solid-enriched sediment to enter preferentially into the tube. The distance at which the grout penetrates the column, and the corresponding time were used to draw a curve to measure of injectability. As it can be seen in Figure 5, the Ledan and Bio Aedilitia grouts were more easily injected than the PLM grouts at w/s of 0.8.

Fig. 3
Injection test results



The grouts with the modified w/s optimised by the rheological experiments described in the following at first seemed to behave better and to be correctly fluid during the injection testing. However, during the setting up of the instrument, it was noticed that the grouts exerted more friction inside the stopper in contact with the crushed brick. This caused an inadequate flow and grout rising in the column. Grout Ledan TB1 injected at w/s 0.7 filled 10 cm of column over 36 cm of total length, grout PLM-1 with w/s 0.7 and grout PLM-SM with w/s 0.5 filled respectively 5 cm and 2 cm of column over 36 cm of total length. Grout Bio Aedilitia did not even made it to the crushed brick column as it formed a plug along the connection tube between the grout reservoir and the beginning of the column. All these tests were carried out without pre-wetting, and it was measured that crushed bricks absorb 28,4% by weight of water. Thus, it was decided to pre-wet the column with an injection of water and let the water drain overnight. The penetration of grout PLM-SM increased from 2 cm without pre-wetting to 8 cm with pre-wetting. Prewetting did not significantly change the injection performance.

4.5 Rheological Measurements

On the two Ledan grouts, a rheological comparison was made between the slurry before injection and after injection. Both materials had a pseudoplastic behaviour (shear thinning). Grout Ledan TA1 showed a similar rheological behaviour before and after injection and was thus very stable during the injection. Given the stability of the sample, it was assumed as the reference of a correct flow behaviour for comparison purposes with the other samples. Its range of shear stresses is marked in yellow in Figure 4. A more pronounced difference was observed in Ledan TB1 which is more fluid before the injection test than after.

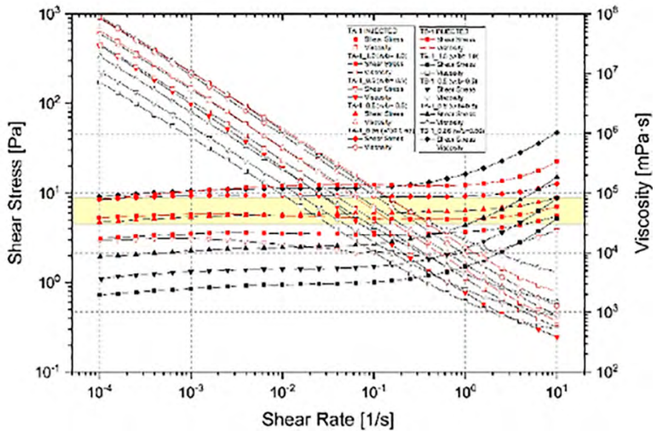


Fig. 4
Summary of the rheometric measurements for Ledan TB1 and Ledan TA1. The yellow band marks the reference TA1 shear stress at w/b 0.8 used as a target to optimize w/b for the other grouts. The green band marks the shear rate experienced during injection.

On the two PLM grouts, rheological comparisons were made between w/s 0.8 and 0.66. These two values were chosen because they are respectively the one used for injection and the one recommended in the data sheets. Both materials in each case have a pseudoplastic behaviour (shear thinning) but at high shear rates viscosity levels off and shear stress increases. Grout PLM-1 at w/s 0.8 has a behaviour similar to grout PLM-SM at w/s 0.66.

It was difficult to perform rheological measurements of grout Bio Aedilitia with a water-cement ratio of 0.8 due to the extremely low viscosity and high sedimentation rate of the material. However, both pre-injection and post-injection measurements were carried out. A very different behaviour was found at w/s 0.3. At low shear rate it was shear thinning. On increasing shear rate the behaviour changes to shear thickening finally reaching higher viscosity than all the other grouts analysed.

4.6 Image analysis of the Bed Porosity

The pore structure infilled by the grouts are presented was analysed by image analysis of polished column sections. The sections were covered with a blue lapis lazuli pigment to identify air bubbles which were often found near the edges of the crushed brick. 3 phases were classified. The crushed bricks representing 56,4% ± 2,2 of the surface, the grout representing 32,4% ± 0,2 and the air bubbles representing 11,2% ± 0,3. The error of the analysis is less than 2%.

From the grout flow rate and the average diameter of the porosity it was possible to estimate that the grout shear rate inside the injected column was included in the range $14 \leq \gamma \leq 320\ s^{-1}$.

4.7 Mechanical Strength and Resistance to Salt Corrosion

The Ledan TA1 and PLM-SM grouts have about twice the compressive strength than the other materials as shown in Figure 5a. Moreover, while for Ledan TA1 the bottom sample has a higher strength than the top sample, the other samples that filled the injection column, Ledan TB1 and Bio Aedilitia, show the opposite.

Fig. 5
Grouts' unconfined compressive strength and indirect tensile (splitting) strength

The results of the indirect tensile strength measurements, shown in Figure 5b, follow the same trends. Similar results are obtained for samples of grouts infilling the crushed brick column and for samples of grout only without crushed bricks.

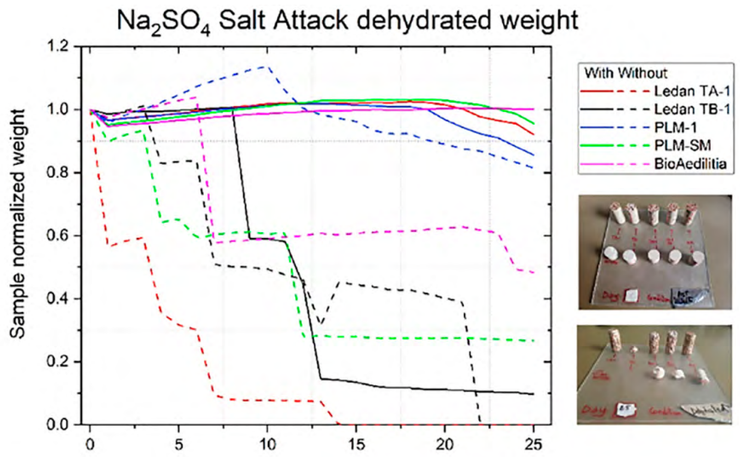


Fig. 6
Sulfate resistance tests result: (a) Dehydrated Weight in 25-Day Na_2SO_4 Salt Attack; (b) Sample Before Sulfate Attack; (c) Sample After Sulfate Attack.

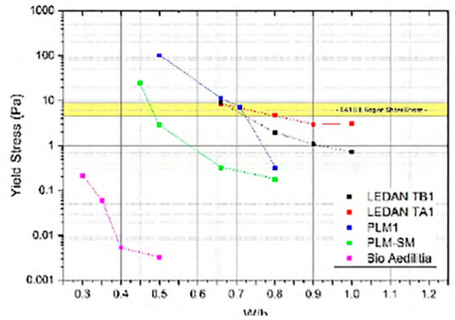
The salt resistance data are reported in Figure 6 as weight measurements after imbibition in the 6 wt% Na_2SO_4 solution. The weights were normalized to the initial samples' weight. All samples have been prepared with w/s 0.8. The samples containing crushed bricks showed a very good resistance to sulfates for the first 17 days, except Ledan TB1 which on the 8th day fractured and divided in two parts. On the seventeenth day, all samples except Bio Aedilitia begun to show corrosion on the external cylinder surface, where the grout was lost. The measurements on the samples without crushed brickshow that the Ledan grouts are not very resistant. Ledan TA1 started to lose considerable side portions of the sample already after the first days. Both samples failed the test not reaching the 25th day. The grout Bio Aedilitia was the best showing only some corrosion on the external cylinder surface and maintaining a compact core until the end of the test.

5 Discussion

5.1 Rheological Behaviour

Except for Bio Aedilitia, all samples display a pseudoplastic shear thinning behavior. Bio Aedilitia, on the other hand, requires less water to achieve the same viscosity at low to intermediate shear rates. At the highest shear rates experienced in the test column, Bio Aedilitia exhibits a shear thickening behavior. The yield stress of the grouts decreases with an increase in w/s, as illustrated in Figure 7. The yellow band indicates the optimal yield stress demonstrated by Ledan TA1, characterized by easy injectability and high stability, where the grout outflows from the test column similar to the grout entering. The optimal grouts exhibit minimal variations in yield stress with changing w/b, particularly pronounced for the grouts with small particle sizes. In contrast, the grouts with larger particles and a steeper relation between yield stress and w/s do not enhance their injectability even when prepared at the w/s corresponding to the optimal yield stress.

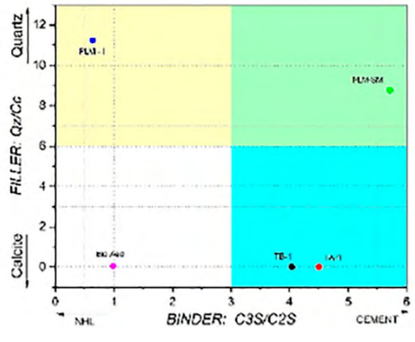
Fig. 7:
Summary of the results of the rheological measurements for w/s optimization.



5.2 Mechanical Strength and Resistance to Sulfates

The difference in mechanical strength and sulfate resistance between the different injection grouts is related to the binder nature as shown in Figure 10. Samples containing NHL as a binder and low C_3S/C_2S values, such as the grout PLM-1, have low compressive strength and moderate resistance to salt attack. Samples showing high C_3S/C_2S values and Portland cement as a binder, such as grout Ledan TA1 and grout PLM-SM have the highest strength but low salt resistance.

Fig. 8
Compositional characteristics of the grouts. Binder (abscissa) and filler (ordinate).



Conclusion

- The mineralogical composition of the grouts has been determined by XRPD and FTIR. The grouts are composed by a binder and a filler. The filler can be either limestone or silica, and it represent slightly more than 50 wt% in all the grouts except in Ledan TB1 having about 75 wt% filler. The binder is based on Portland cement in most of the grouts, except in PLM-1 and in Bio Aedilitia where it is based on NHL.
- The grouts can be divided in two groups concerning the particle size distribution. The Ledan products show a principal mode centred at about 3 µm, while all the others show a principal mode centred at about 20 µm.
- Parallel to the differences in particle size distribution, the grouts show differences in fluidity when mixed with equal water-to-solid ratios. The Ledan grouts are less fluid compared to the others and show a higher yield stress. Having a finer particle size distribution, the Ledan grouts show a higher water demand. And having a higher yield stress, the Ledan grouts are more stable against sedimentation.
- We compared initially the grouts at an equal w/s of 0.8. The different grouts showed different fluidity, for some of them far from the optimal value estimated by the yield stress value of the best performing grout.
- The mechanical strength is related to the nature of the binder. Grouts showing higher C3S/C2S values and probably containing a cement-based binder have the highest mechanical strength. However, these grouts show the lowest resistance to sulphate attack, while the grouts showing low C3S/C2S values and probably containing an NHL binder have the highest resistance to sulphates.
- Optimization of the grouts' fluidity, carried out by selecting the optimum w/s improves the stability of the grouts to sedimentation but does not improve the overall injectability. It is not possible to control viscosity at both low (for sedimentation) and high (for injection) shear rates by only modifying w/s. Moreover, the high-water absorption of the crushed brick may require additional water.

The key to having a grout with good injectability lies in its cohesivity which can be controlled by selecting the proper particle size distribution, or eventually with the addition of rheology modifiers. The sulphate resistance is obtained by the proper choice of the binder, with NHL performing better than cement.

References

1. Biçer-Simsir & Rainer (2011) Evaluation of lime-based hydraulic injection grouts for the conservation of architectural surfaces. A Manual of Laboratory and Field Test Methods. The Getty Conservation Institute
2. Ferragni D, Forti M, Malliet J, Mora P, Teutonico JM, and Torraca G (1984) 'Injection grouting of mural paintings and mosaics', in Adhesives and Consolidants, Preprints of the Contributions to the Paris Congress, 2–8 September 1984, ed. N.S. Brommelle, E.M. Pye, P. Smith and G. Thomson,International Institute for Conservation, London 110–116./
3. Ferragni D, Forti M, Malliet J, Teutonico JM, and Torraca G (1985) 'In situ consolidation of wall and floor mosaics by means of injection grouting techniques', in Mosaics, No.3: Conservation in situ, Aquileia 1983, ICCROM, Rome 83–101.
4. Roby TC (1994) 'Consolidation of a floor mosaic during the excavation of a Byzantine Church in Petra, Jordan', in Fifth Conference of the International Committee for the Conservation of Mosaics: Proceedings, Faro e Conimbriga, 1993, ed. International Committee for the Conservation of Mosaics and Museu Monográfico de Conimbriga, ICCM, Conimbriga 31–37.
5. Demitry L (1994) 'Misure preventive di conservazione in situ di mosaici pavimentali durante uno scavo di emergenza', in Mosaicos, no.5: Conservación in situ: Palencia, 1990: IV Conferencia general del comité internacional de mosaicos, ed. ICCM and ICCROM, Excma, Diputación Provincial, Departamento de Cultura, Palencia 161–171.
6. Binda L, Modena C, Baronio G, and Gelmi A (1994) Experimental qualification of injection admixtures used for repair and strengthening of stone masonry walls. Proceeding of the 10th International Brick and Block Masonry Conference, Calgary, 539–48.
7. Piqué F, Burch R, Roby T, and Demas M (2008) 'Learning from the literature: a review of published works on mosaics deterioration and conservation treatments', in Lessons Learned: Reflecting on the Theory and Practice of Mosaic Conservation. Proceedings of the 9th Conference of the International Committee for the Conservation of Mosaics, Hammamet, Tunisia, Nov 29–Dec 3, 2005, ed. A. Ben Abed, M. Demas and T. Roby, The Getty Conservation Institute, Los Angeles 33–45.
8. Veiga R (2010) Conservation of historic renders and plasters: From laboratory to site. In 2nd conference on historic mortars - HMC 2010 and RILEM TC 203-RHM final workshop, ed. J. Válek, J. J. Hughes, and C. J. W. P. Groot, 1241–56. Prague, Czech Republic: RILEM Publications SARL
9. Zajadacz K, and Simon S (2006) 'Grouting of architectural surfaces: The challenge of testing', in Theory and Practice in Conservation: International Seminar, ed. J. Delgado Rodrigues and J. Mimoso, National Laboratory of Civil Engineering, Lisbon 509–517.
10. Valluzzi MR (2009) 'Requirements for the choice of mortar and grouts for consoli-

dation of three-leaf stone masonry walls', in RILEM International Workshop Repair Mortars for Historic Masonry, Delft University of Technology, The Netherlands, 26–28 January 2005, ed. C. Groot, RILEM Publications SARL 382–397.
11. Baglioni P, Dei L, Piqué F, Sarti G, and Ferroni E (1997) 'New autogenous lime-based grouts used in the conservation of lime-based wall paintings', Studies in Conservation 42(1) 43–54.
12. Griffin I (2004) 'Pozzolanas as additives for grouts: An investigation of their working properties and performance characteristics', Studies in Conservation 49(1) 23–34.
13. Van Rickstal F, Toumbakari EE, Ignoul S, and Van Gemert D (2003)'Development of mineral grouts for consolidation injection', in Consolidation of Masonry, Advances in Materials Science and Restoration, ed. D. Van Gemert, Verlag, Freiburg 1 61–75.
14. Miltiadou-Fezans A (1998) 'Criteria for the design of hydraulic grouts injectable into fine cracks and evaluation of their efficiency', in Compatible Materials for the Protection of European Cultural Heritage, ed. G. Biscontin, A.I. Moropoulou, M. Erdick and J. Delgado Rodrigues, Technical Chamber of Commerce, Athens PACT 56 149–163.
15. Klarner K, and Schick B (1998) 'Foamgrout: A method to treat large detachments of plaster layers carrying wallpaintings', in Baroque Wallpaintings: Technical Problems and Current Conservation Methods, ed. G. Krist and S. Winkelbauer, Bundesdenkmalamt, Wien, Austria 117–118.
16. Keersmaekers R, Schueremans L, Van Rickstal D, Van Gemert M, Knapen D, and Posen D (2007) 'NTDControl of injection of an appropriate grout mixture for the consolidation of the columns foundations of Our Lady's Basilica at Tongreren (B)', in Proceedings of the 5th International Conference [on] Structural Analysis of Historical Constructions: Possibilities of Numerical and Experimental Techniques, 6–8 November 2006, ed. B.P. Lourenço, P. Roca, C. Modena and S. Agrawal, Macmillan, New Delhi 553–560.
17. Ignoul S, Van Gemert D, and Van Rickstal F (2003) 'Application of mineral grouts for structural consolidation of historical monuments', Internationale Zeitschrift für Bauinstandsetzen 9(4) 365–382.
18. Rousset B, Gentile S, James J, and Pozzi B (2018) 'Injection grouts for molasse sandstones: preliminary assessment'
19. Sharma R, Gupta H, and Kanotra Y (1995) 'Evaluation of physical characteristics of "Mercula" admixed clay for sealing of cracks in mud plaster of monasteries, Ladakh Region', in Methods of Evaluating Products for the Conservation of Porous Building Materials in Monuments, International Colloquium, Rome, 19–21 June 1995, ICCROM, Rome 41–48.
20. Toumbakari EE, and Van Gemert D (1998) 'Injection grouts for ancient masonry: strength properties and microstructural evidence', in Compatible Materials for the Protection of European Cultural Heritage, ed. G. Biscontin, A.I. Moropoulou, M. Erdik and J. Delgado Rodrigues, Technical Chamber of Commerce, Athens PACT 56 191–200.
21. Penelis G, Karaveziroglou M, and Papayianni J (1989) 'Grouts for repairing and strengthening old masonry', in Structural Repair and Maintenance of Historical Buildings, ed. C.A. Brebbia, Computational Mechanics Publications, Southampton 179–188.
22. Egloffstein P, and Kenner C (2006) 'Hinterfüllmörtel für die Konservierung mittelalterlicher Verputze', in Restaurierungsmötel in der Denkmalpflege, Monudocthema 03, ed. E. Koser, Fraunhofer IRB Verlag, Stuttgart 178–199.
23. Flatt RJ, and Girardet FJ (2000) 'Injectable slurries for the in-situ conservation of pavement mosaics', in Proceedings of the 9th International Congress on Deterioration and Conservation of Stone, Venice, June 19–24, 2000, ed. Vasco Fassina, Elsevier, Amsterdam Vol.II 297–305.
24. Dominik A, Meyer U, and Schubert P (1994) 'Entwicklung von Schlämmen und Verfugmörteln zur Instandsetzung der Außenfassaden der Matthiaskapelle Kobern-Gondorf ', in Jahresberichte aus dem Forschungsprogramm Steinzerfall, Steinkonservierung: ein Förderprojekt des Bundesministers für Forschung und Technologie, ed. R. Snethlage, Ernst & Sohn, Berlin 135–144.
25. Böttger KG, and Knöfel D (1993) 'Development and testing of injection mortars of high sulfate resistance', in Proceedings of the International RILEM/UNESCO Congress 'Conservation of Stone and Other Materials: Research-Industry-Media' held at the UNESCO Headquarters, Paris, June 29 – July 1, 1993, ed. M.J. Thiel, E & F.N. Spon, London Vol. II 629–636.